Supporting Information

Title: A practical carbon model for managing peri-urban bamboo forests as nature-based climate solutions

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Method S1. Details of a partial equilibrium model for carbon allocation in forest ecosystem _ annual dynamic (PEMCAFE_ad)

The PEMCAFE_ad, in its current iteration (annual dynamic version 0.8, dated 30 August 2024), represents a significant advancement in easily accessing data *in situ*[1] and focusing on quantifying carbon dynamics within terrestrial ecosystems. The version was expanded from the basics of an input-output analysis (Leontief, 1963) combined with the concept of process-based, carbon budget, and gain-loss models (Running & Coughlan, 1988; Harmon et al., 1990; Dixon et al., 1994)[2].

At its core, PEMCAFE_ad employs a series of interconnected equations to estimate aboveground net primary productivity (ANPP) and belowground net primary productivity (BNPP). The ANPP calculation ingeniously incorporates not only changes in aboveground carbon stocks but also litterfall contributions[3], the BNPP of each component, and soil carbon dynamics, with the latter adaptable to scenarios involving wood or culm harvesting. Then, endogenously solve to optimise the turnover rate of each organ for the following simulations.

The model's approach to BNPP estimation is particularly noteworthy, offering two methodologies. The first method combines the change in belowground carbon with belowground detritus, while the second introduces soil autotrophic respiration as a key component. This dual approach allows for flexibility in application, depending on the specific research context and data availability.

The PEMCAFE_ad has 10 initialisation and 4 constraint input variables (Table S1) and 46 equations and inequations (Table S2). Based on the assumption of total gross

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[1] This allows for a very simple annual survey of aboveground biomass in a reasonable cost to accurately estimate the entire ecosystem carbon flux after treatments, even without setting up carbon flux towers and using the eddy-covariance (EC) method. The motivation for this idea stems from the author's firsthand experience at UNFCCC conferences, where representatives from developing and developed countries were locked in heated debates over long-term monitoring funding.

The representatives from developing nations argued that developed countries should bear the greatest responsibility for climate change due to their historical emissions and should cover the costs of any investigations and assessments. On the other hand, representatives from developed nations contended that funding granted to developing countries often falls prey to corruption, with a shortage of qualified researchers to carry out the necessary work. They also highlighted the practical challenges of implementing long-term fundamental scientific monitoring, let alone conducting mitigation and adaptation policy simulations and assessments.

However, all nations, whether developing or developed, are on the same Noah's Ark, facing the intensification of extreme climate events in the future, and should collaborate to find solutions. Problems should be addressed, not ignored.

To overcome this challenge, the authors of this paper have attempted to create a monitoring system with very low costs and technical barriers. The method only requires a functioning computer running python codes, alongside annual measurements of DBH in plants and average temperature data. This approach eliminates the need for continuous maintenance of carbon flux towers, while achieving validation results comparable to those of more expensive EC methods.

[2] The method here need to satisfy two conditions, (1) the fact that leaf net productivity (LNP) and BNPP have a strong nonlinear correlation (It was satisfied in *P. edulis* forests, Wang et al., 2023) and (2) the optimisation of the turnover rate and decomposition rate of all components in the target terrestrial ecosystem fitted by aboveground dynamics as constraints concisely reflects the heterotrophic respiration and coarse woody detritus (CWDs) without drastic lateral movement.

[3] Here is including coarse woody detritus-related carbon which highly correlated with heterotrophic respiration in aboveground neglected in a lot of previous studies due to the difficulty of measuring and the complexity of various distribution in time-space (Harmon et al., 2011).

photosynthesis (input) (G_p) equal to total autotrophic respiration ($R_{A\ total}$) plus total net growth increment (G_n), total litterfall (L_t), and belowground detritus (D_{below}) (output) (in short term: input = output) as below:

$$G_p = R_{A\ total} + G_n + L_t + D_{below} \qquad (EqMS\ 1)$$

If G_n presents in carbon equivalent, it is defined as the same as the total carbon sink (ΔTotal, Mg C ha^{-1} yr^{-1}) here as $EqMS$ 2:

$$G_p = R_{A\ total} + \Delta\text{Total} + L_t + D_{below} \qquad (EqMS\ 2)$$

where $R_{A\ total}$ can be divided by aboveground autotrophic respiration ($R_{A\ above}$) plus belowground autotrophic respiration ($R_{A\ below}$) as $EqMS$ 3:

$$R_{A\ total} = R_{A\ above} + R_{A\ below} \qquad (EqMS\ 3)$$

ΔTotal can be divided by aboveground carbon sink (ΔAbove) and belowground system carbon sink (ΔBelow) as $EqMS$ 4:

$$\Delta\text{Total} = \Delta\text{Above} + \Delta\text{Below} \qquad (EqMS\ 4)$$

Substitute $EqMS$ 3 and $EqMS$ 4 into $EqMS$ 2, can be obtained as $EqMS$ 5:

$$G_p = \Delta\text{Above} + \Delta\text{Below} + R_{A\ above} + R_{A\ below} + L_t + D_{below} \qquad (EqMS\ 5)$$

If all components are calculated by carbon equivalent, they also can be presented as below, which is the familiar equation for whom studied in forest ecosystems:

$$\boldsymbol{G_p = \text{ANPP} + \text{BNPP} + R_{A\ above} + R_{A\ below}} \qquad (EqMS\ 6)$$

where ANPP means aboveground net primary productivity and BNPP means belowground net primary productivity.

The ANPP is consist of leaf net productivity (LNP), branch net productivity (BNP), culm net productivity (CNP), litterfall of leaf (LD), litterfall of branch (BD), litterfall of culm (CD), and litterfall of others ($Dothers$). It can be described as $EqMS$ 7:

$$\begin{aligned}\text{ANPP} &= \Delta\text{Above} + L_t \\ &= LNP + BNP + CNP \\ &\quad + LD + BD + CD \qquad (EqMS\ 7)\end{aligned}$$

where ΔAbove means total aboveground carbon sink, and L_{total} means total litterfall.

The BNPP is consist of root net productivity ($RoNP$), rhizome net productivity ($RhNP$), stump net productivity ($StNP$), detritus of roots (RoD), rhizomes (RhD), and stumps (StD), and described as $EqMS$ 8:

$$\text{BNPP} = \Delta\text{Below} + D_{\text{below}}$$

$$= RoNP + RhNP + StNP + RoD + RhD + StD \qquad (EqMS\ 8)$$

where ΔBelow means total belowground carbon sink.

The $R_{A\ above}$ can be calculated by summing up leaf respiration (R_L), branch respiration (R_B), culm respiration (R_C), and any other organ respiration (R_{others}) and presented as $EqMS$ 9[4]:

$$R_{A\ above} = R_L + R_B + R_C + R_{others} \qquad (EqMS\ 9)$$

(In tree species, R_C is commonly used by stem or truck respiration [R_{Stem}], and $R_L + R_B$ is presented as foliage respiration [$R_{Foliage}$] depending on cases.)

The $R_{A\ below}$ can be calculated by summing up root respiration (R_{Ro}), rhizome respiration (R_{Rh}), and stump respiration (R_{St}) presented as $EqMS$ 10:

$$R_{A\ below} = R_{Ro} + R_{Rh} + R_{St} \qquad (EqMS\ 10)$$

here, the rhizome is a specified organ for *P. edulis* and some other bamboo or grass species. Tree species can replace the organ to course roots if they do not have them.

In fact, the amount of D_{below} is difficult to estimate. Here, we employed the method that optimises the turnover rate of each organ, that is, the net carbon changes in soil (ΔSoil) is equal to detritus import to soil minus the $R_{A\ below}$ and respiration of mineral soil with organic matters ($R_{H\ soil}$), and can be described as below:

$$\Delta\text{Soil} = D_{below} + D_{Litter_layer} - R_{H\ soil}$$

that is,

$$D_{below} = R_{H\ soil} - D_{Litte_layer} - \Delta\text{Soil} \qquad (EqMS\ 11)$$

where $R_{H\ soil}$ means heterotrophic respiration of mineral soil with organic matters, and D_{Litter_layer} means detritus imported from litter layer to the soil.

Based on the derivation and description of the above equations, the turnover of each organ ($D_{turnover\ of\ each\ organ}$) is a linchpin for estimating the dynamics of carbon allocation in the further period[5], and can be derived as below:

$$D_{turnover\ of\ each\ organ} = \frac{D_{each\ organ}}{Biomass_{each\ organ} + \Delta Biomass_{each\ organ}} \qquad (EqMS\ 12)$$

[4] In tree species, R_C is commonly used by stem or truck respiration [R_{Stem}], and $R_L + R_B$ is presented as foliage respiration [$R_{Foliage}$] depending on cases.

[5] Through the method of endogenously solving the turnover rate of each organ, underground carbon storage and soil carbon dynamics can be cleverly estimated, even though it was considered difficult to describe and quantify in the past (Keith et al., 2009)

where $Biomass_{each\ organ}$ can be represented as the carbon stock of each organ (e.g. foliage, branches, culms, stumps, rhizomes, roots) and sum up of $Biomass_{each\ organ}$ will get total biomass ($Biomass$) as follows:

$$\begin{aligned} Biomass &= \sum Biomass_{each\ organ} \\ &= Foliage + Branches + Culms + \\ &\quad Stumps + Rhizomes + Roots \\ &= AGC + BGC \end{aligned} \qquad (EqMS\ 13)$$

ΔB$iomass$, the carbon sink of total organs (net productivity of total organs), is not easy to obtain reliable data in the field that is affected by the way of investigation, especially in the belowground. So here, we employed linear and non-linear models obtained from Wang et al., 2023 to describe the relationships between *StNP-CNP*, *RhNP-LNP*, and *RoNP-RhNP* (shown in Table S2). It is also possible to apply the IPCC's method (*BGC*/*AGC* ratio equal to $\Delta Below/\Delta Above$) if the tree or plant species is not so complicated as *P. edulis* and validated by worldwide data:

$$BGC = Roots + Rhizomes + Stumps \qquad (EqMS\ 14)$$

$$\begin{aligned} \Delta Below &= RoNP + RhNP + StNP \\ &= BGC_{(t)} - BGC_{(t-1)} \end{aligned} \qquad (EqMS\ 15)$$

where the BGC means belowground carbon stocks, the $\Delta Below$, $RoNP$, $RhNP$, and $StNP$ means belowground, roots, rhizomes, and stump carbon sink, $respectively$; AGC means aboveground carbon stock, and can be summed up by leaf carbon stock ($Foliage$), branch carbon stock ($Branches$), and culm carbon stock ($Culms$), as follows:

$$AGC = Foliage + Branches + Culms \qquad (EqMS\ 16)$$

ΔAbove, the aboveground carbon sink, can be summed up by leaf carbon sink (LNP), branch carbon sink (BNP), and culm carbon sink (CNP), as follow:

$$\begin{aligned} \Delta\text{Above} &= LNP + BNP + CNP \\ &= AGC_{(t)} - AGC_{(t-1)} \end{aligned} \qquad (EqMS\ 17)$$

After the journey of the derivation of the above equations, we also need to sort out the equation of $Litter\ layer$. From the method of input equal to output, we conduct that:

$$\begin{aligned} Litter\ layer &= LD + BD + CD + otherD \\ &= D_{Litter\ layer} + \Delta Litter\ layer + R_{H\ Litter\ layer} \end{aligned}$$

that is,

$$\begin{aligned} \Delta Litter\ layer &= Litter\ layer - R_{H\ Litter\ layer} \\ &\quad - D_{Litter\ layer} \end{aligned} \qquad (EqMS\ 18)$$

where $\Delta Litter\ layer$ means carbon sinks in $Litter\ layer$, and $R_{H\ Litter\ layer}$ means heterotrophic respiration of $Litter\ layer$[6].

Commonly, $R_{H\ Litter\ layer}$ is measured by soil respiration with $Litter\ layer$ ($R_{with\ Litter\ layer}$) minus soil respiration without $Litter\ layer$ ($R_{without\ Litter\ layer}$):

$$\begin{aligned} R_{H\ Litter\ layer} &= R_{with\ Litter\ layer} - R_{without\ Litter\ layer} \\ &= R_{with\ Litter\ layer} - (R_{A\ below} + R_{H\ soil}) \end{aligned} \qquad (EqMS\ 19)$$

However, the heterotrophic respiration of $Litter\ layer$ in a lot of cases neglected the coarse wood detritus (CWDs). Here, we considered the emission and estimated them by the constraints of aboveground carbon stock data and fitted the turnover rate of each organ to estimate the net ecological productivity (*NEP*) elegantly by minimising the root mean square error (RMSE) between the next time state of total ecological carbon stocks (TEC_{t+1}) minus initial state one (TEC_t) (also known as ΔTEC) and the total net primary productivity (*TNPP*) minus R_H.

$$\begin{aligned} \Delta TEC &= TNPP - R_H \\ &= ANPP + BNPP - R_{H\ Litter\ layer} - R_{H\ Soil} \end{aligned} \qquad (EqMS\ 20)$$

The above simultaneous equations of the PEMCAFE describe the basic prototype of the input-output of the forest ecosystem (Table S2).

Monte Carlo uncertainty propagation

To quantify predictive uncertainty arising from measurement error in input observations, PEMCAFE employs a Monte Carlo propagation procedure. In each of *n* simulations (default 1,000; 10,000 used in this study), the six organ-specific biomass input variables are independently perturbed by additive Gaussian noise, while the endogenously optimised turnover parameters (Table S1) are held fixed at their best-fit values. The standard deviations used for perturbation are listed in Table SM1 and reflect typical field-measurement variability for each organ. Perturbed biomass values are constrained to remain non-negative (minimum 0.01 Mg C ha^{-1}). For each perturbed input dataset, the model is run forward using the fixed parameter set, yielding one complete set of output variables. The 95% confidence intervals are then derived from the t-distribution of the Monte Carlo replicates.

This design isolates the effect of input-observation uncertainty on model outputs. Because PEMCAFE's turnover parameters are jointly constrained by the equilibrium closure condition (NEP from ΔTEC = NEP from TNPP − R_H), independent perturbation of individual parameters would violate the internal consistency of the framework. Holding parameters fixed therefore represents the appropriate uncertainty propagation strategy for a system of jointly determined variables.

[6] If possible, this could even be subdivided into standing dead (the part not touching the soil) culm litter. This may result in different rates of decomposition and heterotrophic respiration (Orrego et al., 2023). This issue often occurs in the real world and is a critical consideration, yet it is almost always overlooked by models.

Table SM1. Default standard deviations (Mg C ha^{-1}) used for Monte Carlo input perturbation.

Input variable	SD (Mg C ha^{-1})
Foliages	0.3
Branches	0.7
Culms	3.2
Roots	0.4
Rhizomes	0.3
Stumps	1.1

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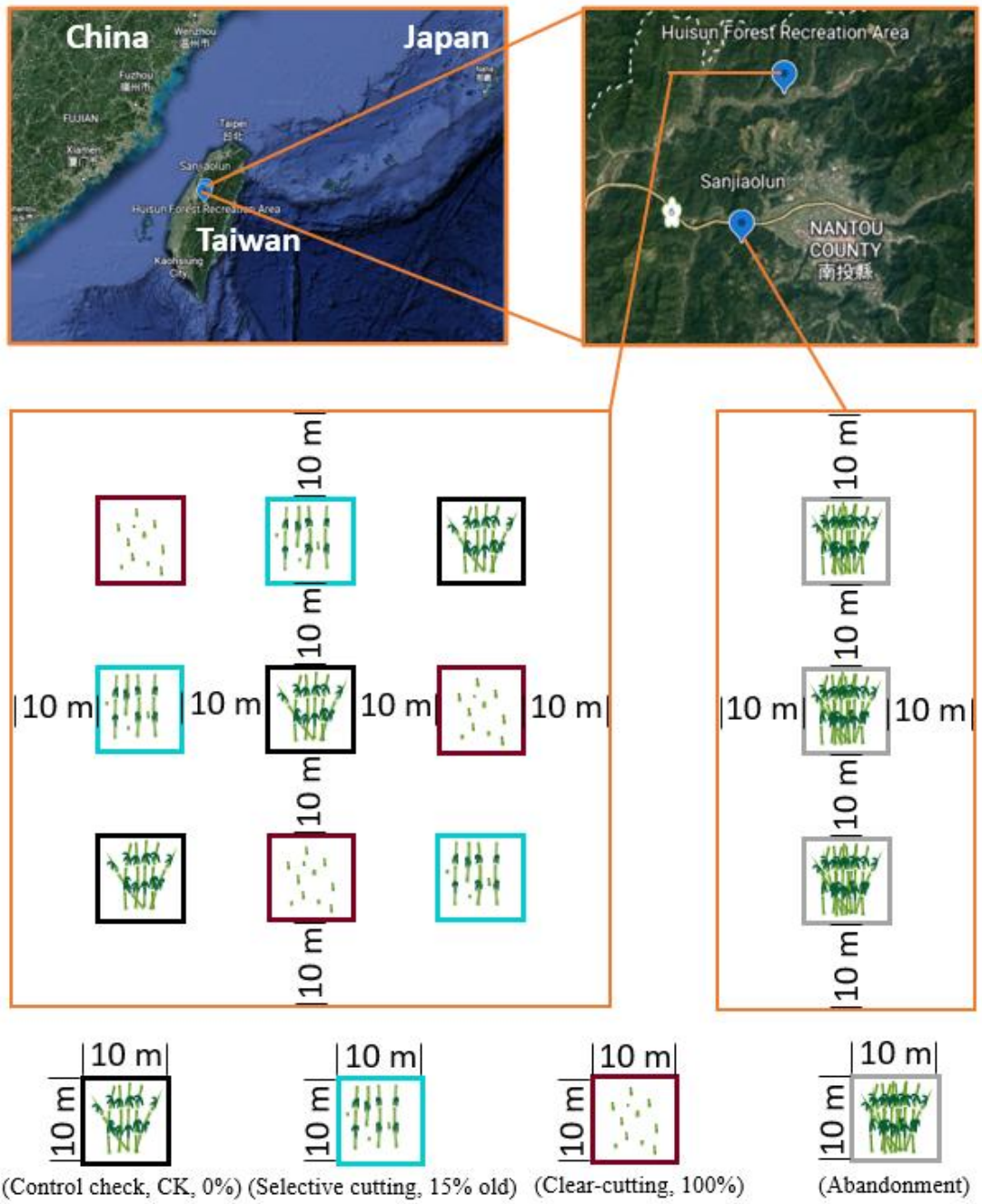

Figure S1. Distribution diagram of experimental sample plots. The top panels show the location of the study site within Taiwan, specifically the Huisun Forest Recreation Area in Nantou County. The bottom panels illustrate the layout of the experimental plots. Black plots represent control (CK, 0% cutting). Dark turquoise plots indicate selective cutting (SC, 15-30% of old culms removed). Burgundy red plots represent clear-cutting (CC, 100% cutting). Grey plots denote abandonment (AB) after cessation of farmer management practices. All plots are 10 m×10 m in size and are separated by a 10 m buffer zone. This figure was generated using Google Earth (2025).

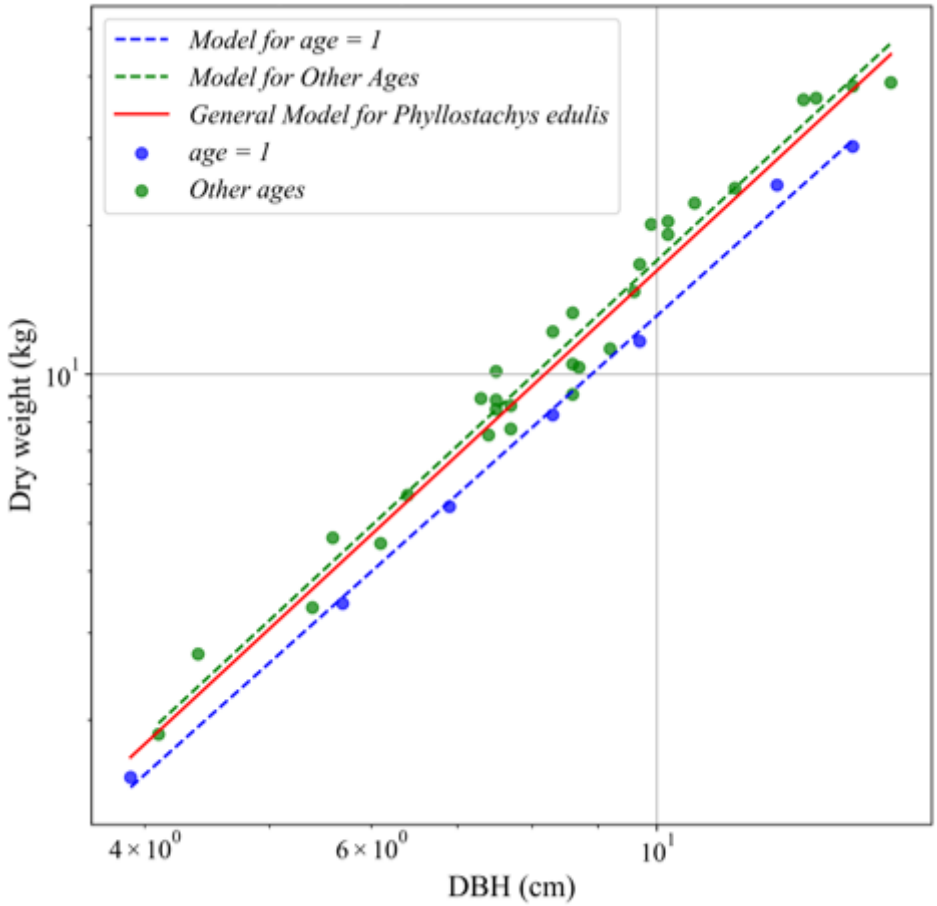

Figure S2. Allometric models for *Phyllostachys edulis* dry weight prediction. The figure presents three distinct models: a specific model for 1-year-old bamboo (blue dashed line), a model for bamboo of other ages (green dashed line), and a general model applicable to all P. edulis (solid red line). Corresponding data points are plotted for 1-year-old specimens (blue circles) and specimens of other ages (green circles). Both axes are on a logarithmic scale.

Table S1. Variables and parameters in PEMCAFE (annual dynamic version)

Category	Variables	Units	Definition
Input (initialising, t=0)	AvgTemp	˚C	Average of mean daily temperatures in a year. (Some of research use the Average of mean monthly temperatures in a year depending on their definition of weather stations.)
Input (initialising, t=0)	Foliages	Mg C ha^{-1}	The leaf biomass (carbon equivalent).
Input (initialising, t=0)	Branches	Mg C ha^{-1}	The branch biomass (carbon equivalent).
Input (initialising, t=0)	Culms	Mg C ha^{-1}	The culm (or the truck in tree species) biomass (carbon equivalent).
Input (initialising, t=0)	Roots	Mg C ha^{-1}	The root biomass (carbon equivalent).
Input (initialising, t=0)	Rhizomes	Mg C ha^{-1}	The rhizome (or the coarse root in tree species) biomass (carbon equivalent).
Input (initialising, t=0)	Stumps	Mg C ha^{-1}	The stump biomass (carbon equivalent).
Input (initialising, t=0)	Litter_layer	Mg C ha^{-1}	The litter mass (carbon equivalent).
Input (initialising, t=0)	SC	Mg C ha^{-1}	The soil carbon storage.
Input (initialising, t=0, optional)	Undergrowth	Mg C ha^{-1}	The undergrowth biomass (carbon equivalent).
Input (constraint, t>0)	Foliages	Mg C ha^{-1}	The leaf biomass (carbon equivalent) of the following investigation.
Input (constraint, t>0)	Branches	Mg C ha^{-1}	The branch biomass (carbon equivalent) of the following investigation.
Input (constraint, t>0)	Culms	Mg C ha^{-1}	The culm biomass (carbon equivalent) of the following investigation.
Input (constraint, t>0, optional)	Undergrowth	Mg C ha^{-1}	The undergrowth biomass (carbon equivalent) of the folowing investigation.
Output	AGC	Mg C ha^{-1}	The aboveground biomass (carbon equivalent).
Output	Root_Shoot_Ratio		The ratio of belowground divided by aboveground biomass (carbon equivalent).
Output	BGC	Mg C ha^{-1}	The belowground biomass (carbon equivalent).
Output	TC	Mg C ha^{-1}	The toral biomass (carbon equivalent).
Output	TEC	Mg C ha^{-1}	The total ecosystem carbon which includes above- and below-ground live components, dead biomass, and soil carbon.
Output	LNP	Mg C ha^{-1} yr^{-1}	The leaf net primary productivity.
Output	BNP	Mg C ha^{-1} yr^{-1}	The branch net primary productivity.
Output	CNP	Mg C ha^{-1} yr^{-1}	The culm net primary productivity.
Output	LD	Mg C ha^{-1} yr^{-1}	The leaf detritus (carbon equivalent).

Category	Variables	Units	Definition
Output	BD	Mg C ha^{-1} yr^{-1}	The branch detritus (carbon equivalent).
Output	CD	Mg C ha^{-1} yr^{-1}	The culm detritus (carbon equivalent).
Output	Litterfall	Mg C ha^{-1} yr^{-1}	The litter fall (carbon equivalent). (Here is equal to LD+BD+CD)
Output	ANPP	Mg C ha^{-1} yr^{-1}	The aboveground net primary productivity. (Here is equal to LNP+BNP+CNP+ litter fall)
Output	RoNP	Mg C ha^{-1} yr^{-1}	The root net primary productivity.
Output	RhNP	Mg C ha^{-1} yr^{-1}	The rhizome net primary productivity.
Output	StNP	Mg C ha^{-1} yr^{-1}	The stump net primary productivity.
Output	RoD	Mg C ha^{-1} yr^{-1}	The root detritus (carbon equivalent).
Output	RhD	Mg C ha^{-1} yr^{-1}	The rhizome detritus (carbon equivalent).
Output	StD	Mg C ha^{-1} yr^{-1}	The stump detritus (carbon equivalent).
Output	Dbelow	Mg C ha^{-1} yr^{-1}	The total detritus in belowground (carbon equivalent). (Here is equal to RoD+RhD+StD)
Output	BNPP	Mg C ha^{-1} yr^{-1}	The belowground net primary productivity. (Here is equal to equal to RoNP+RhNP+SNP+Dbelow)
Output	TNPP	Mg C ha^{-1} yr^{-1}	The total net primary productivity (Here is equal to ANPP+BNPP)
Output	SR	Mg C ha^{-1} yr^{-1}	The respiration of soils. (Here is equal to Soil_HR+Soil_AR)
Output	Litter_layer_HR	Mg C ha^{-1} yr^{-1}	The heterotrophic respiration in litter layers.
Output	Soil_HR	Mg C ha^{-1} yr^{-1}	The heterotrophic respiration in soil.
Output	HR	Mg C ha^{-1} yr^{-1}	The total heterotrophic respiration. (Here is equal to Litter_layer_HR+Soil_HR)
Output	Foliages_AR	Mg C ha^{-1} yr^{-1}	The autotrophic respiration in leaves.
Output	Branches_AR	Mg C ha^{-1} yr^{-1}	The autotrophic respiration in branches.
Output	Culms_AR	Mg C ha^{-1} yr^{-1}	The autotrophic respiration in culms.
Output	Aboveground_AR	Mg C ha^{-1} yr^{-1}	The autotrophic respiration in aboveground. (Here is equal to Foliages_AR+Branches_AR+Culms_AR)
Output	Soil_AR	Mg C ha^{-1} yr^{-1}	The autotrophic respiration in belowground (soil). (Here is including autotrophic respiration from roots, rhizomes, and stumps)
Output	AR	Mg C ha^{-1} yr^{-1}	The total autotrophic respiration. (Here is equal to Aboveground_AR+Soil_AR)
Output	NEP	Mg C ha^{-1} yr^{-1}	The net ecosystem productivity.

Category	Variables	Units	Definition
Output	DLitter_layer	Mg C ha^{-1} yr^{-1}	The total detritus decomposed from Litter_layer and import to soil (carbon equivalent). (Here is equal to RoD+RhD+StD)
Output	dSC	Mg C ha^{-1} yr^{-1}	The total carbon flux in soil.
Output	GPP (Gp)	Mg C ha^{-1} yr^{-1}	The gross primary productivity. (Here is equal to TNPP+AR)
Automatically fitted parameter	kLitter	yr^{-1}	The decomposition constant in Litter.
Automatically fitted parameter	LTurnoverR	yr^{-1}	The turnover rate in leaves.
Automatically fitted parameter	BTurnoverR	yr^{-1}	The turnover rate in branches.
Automatically fitted parameter	CTurnoverR	yr^{-1}	The turnover rate in culms.
Automatically fitted parameter	StTurnoverR	yr^{-1}	The turnover rate in stumps.
Automatically fitted parameter	RhTurnoverR	yr^{-1}	The turnover rate in rhizomes.
Automatically fitted parameter	RoTurnoverR	yr^{-1}	The turnover rate in roots.
Automatically fitted parameter	Ratio_Litter_layer		

Table S2. Simultaneous equations in the PEMCAFE (annual dynamic version)

equations		notes
$LNP_{(t)}$=	$Foliages_{(t)}$-$Foliages_{(t-1)}$	
$BNP_{(t)}$=	$Branches_{(t)}$-$Branches_{(t-1)}$	
$CNP_{(t)}$=	$Culms_{(t)}$-$Culms_{(t-1)}$	
$AGC_{(t)}$=	$Foliages_{(t)}$+$Branches_{(t)}$+$Culms_{(t)}$	
$StNP_{(t)}$=	$0.1955 \times CNP_{(t)}$	
$RhNP_{(t)}$=	$1.1162 \times LNP_{(t)}^{0.7279}$, if $LNP_{(t)}>0$ 0, if $LNP_{(t)} \leq 0$	
$RoNP_{(t)}$=	$0.9847 \times RhNP_{(t)}$	
$Stumps_{(t)}$=	$Stumps_{(t-1)} + StNP_{(t)}$	
$Rhizomes_{(t)}$=	$Rhizomes_{(t-1)} + RhNP_{(t)}$	
$Roots_{(t)}$=	$Roots_{(t-1)} + RoNP_{(t)}$	
$BGC_{(t)}$=	$Roots_{(t)}$+$Rhizomes_{(t)}$+$Stumps_{(t)}$	
$Root_Shoot_Ratio_{(t)}$=	$BGC_{(t)}/AGC_{(t)}$	
$TC_{(t)}$=	$AGC_{(t)}$+$BGC_{(t)}$	
$LD_{(t)}$=	$Foliages_{(t-1)} \times LTurnoverR$, if $Foliages_{(t-1)}$ is not None, 0, if else	
$BD_{(t)}$=	$Branches_{(t-1)} \times BTurnoverR$, if $Branches_{(t-1)}$ is not None, 0, if else	
$CD_{(t)}$=	$Culms_{(t-1)} \times CTurnoverR$, if $Culms_{(t-1)}$ is not None, 0, if else	
$Litterfall_{(t)}$=	$LD_{(t)} + BD_{(t)}$, if HBP=1, $LD_{(t)} + BD_{(t)} + CD_{(t)}$, if HBP=0	HBP here is a synonym of HWP in tree species
$ANPP_{(t)}$=	$LNP_{(t)}$+$BNP_{(t)}$+$CNP_{(t)}$+$Litterfall_{(t)}$	ANPP here means *in situ*
$StD_{(t)}$=	$Stump_{(t-1)} \times StTurnoverR$, if $Stump_{(t-1)}$ is not None, 0, if else	

equations		notes
$RhD_{(t)}=$	$Rhizome_{(t-1)} \times RhTurnoverR$, if $Rhizome_{(t-1)}$ is not None, 0, if else	
$RoD_{(t)}=$	$Roots_{(t-1)} \times RoTurnoverR$, if $Roots_{(t-1)}$ is not None, 0, if else	
$Dbelow_{(t)}=$	$StD_{(t)}+RhD_{(t)}+RoD_{(t)}$	
$Soil_AR_{(t)}=$	$0.000006 \times BGC_{(t)}^{\ 3.3249}$	It can be devided into Roots, Rhizomes, and Stumps if the fraction of proteins is obtained
$BNPP_{(t)}=$	$StNP_{(t)} + RhNP_{(t)} + RoNP_{(t)} + Dbelow_{(t)}$, if BNPPmethod=1, $StNP_{(t)} + RhNP_{(t)} + RoNP_{(t)} + Soil_AR_{(t)}$, if BNPPmethod=0	
$TNPP_{(t)}=$	$ANPP_{(t)}+BNPP_{(t)}$	
$Soil_HR_{(t)}=$	$0.0071 * ANPP_{(t)}^{3.0772}$, if 4.17<ANPP<11.8, ANPP=4.17, if ANPP<4.17, ANPP=11.8, if ANPP>11.8	4.17<ANPP<11.8 is the observation zone worldwide for *P. edulis* based on collected studies in Wang et al., 2023.
$Foliages_AR_{(t)}=$	$0.1445 \times e^{0.07918AvgTemp(t)} \times 365 \times 24 \times (Foliage_{(t)} / 0.4544) / 1000 * 12 / 44.01$	The Foliages_AR is conducted from daily modules. However, if the data is tough to access, annual AvgTemp is also accaptable to use with low bias.
$Branches_AR_{(t)}=$	$0.215/1.172 \times 0.1445 \times e^{0.07918AvgTemp(t)} \times 365 \times 24 \times (Branches_{(t)} / 0.4544) / 1000 * 12 / 44.01$	The Branches_AR is conducted from daily modules. However, if the data is tough to access, annual AvgTemp is also accaptable to use with low bias.
$Culms_AR_{(t)}=$	$0.085/1.172 \times 0.1445 \times e^{0.07918AvgTemp(t)} \times 365 \times 24 \times (Culms_{(t)} / 0.4544) / 1000 * 12 / 44.01$	The Culms_AR is conducted from daily modules. However, if the data is tough to access, annual AvgTemp is also accaptable to use with low bias.
$Aboveground_AR_{(t)}=$	$Foliages_AR_{(t)} + Branches_AR_{(t)} + Culms_AR_{(t)}$	
$AR_{(t)}=$	$Aboveground_AR_{(t)} + Soil_AR_{(t)}$	
$SR_{(t)}=$	$Soil_AR_{(t)} + Soil_HR_{(t)}$	

equations		notes
$NEP_with_Aboveground_Detritus_Litter_layer_HR_{(t)}=$	$TNPP_{(t)} - Soil_HR_{(t)}$	
$Litter_layer_{(t)}=$	$Litter_layer_{(t-1)} + Litterfall_{(t)} \times kLitter$	
$DLitter_layer_{(t)}=$	$Litter_layer_{(t)} \times kLitter$	
$Litter_layer_HR_{(t)}=$	$Litter_layer_{(t)} \times Rratio_Litter_layer$	
$HR_{(t)}=$	$Soil_HR_{(t)} + Litter_layer_HR_{(t)}$	
$NEP_{(t)}=$	$NEP_with_Aboveground_Detritus_Litter_layer_HR_{(t)} - Litter_layer_HR_{(t)}$	
$SC_{(t)}=$	$SC_{(t-1)} + Dbelow_{(t)} - Soil_HR_{(t)} + DLitter_layer_{(t)}$	
$dSC_{(t)}=$	$SC_{(t)} - SC_{(t-1)}$	
$TEC_{(t)}=$	$TC_{(t)} + Litter_layer_{(t)} + SC_{(t)} + Undergrowth_{(t)}$	
$NEP_from_dTEC_{(t)}=$	$TEC_{(t)} - TEC_{(t-1)}$	
$GPP_{(t)}=$	$TNPP_{(t)} + AR_{(t)}$	
constraints		
LTurnoverR > BTurnoverR		
BTurnoverR > CTurnoverR		
RoTurnoverR > RhTurnoverR		
closure condition		
$NEP_from_dTEC_{(t)}=$	$NEP_{(t)}$	

Table S3. Fundamental characteristics of the bamboo forests under the four treatment types.

Treatments [1]	Culm density (culms ha^{-1})				Average DBH (cm)				Aboveground carbon stocks (Mg C ha^{-1})			
	Ab	CK	SC	CC	Ab	CK	SC	CC	Ab	CK	SC	CC
2008 (before treatment)	5733 (896)	8867 (567)[2]	7400 (1039)	7600 (985)	6.3 (0.2)	6.4 (0.3)	6.4 (0.4)	6.8 (0.2)	38.4 (6.3)	23.6 (1.7)	23.4 (6.9)	32.2 (3.2)
2008 (after treatment)	5733 (896)	8867 (567)	5533 (1250)	0 (0)	6.3 (0.2)	6.4 (0.3)	6.7 (0.5)	n.a	38.4 (6.3)	23.6 (1.7)	20.1 (7.2)	0 (0)
2009	7100 (1058)	10833 (2386)	9933 (1258)	2600 (624)	6.2 (0.2)	5.9 (0.3)	5.9 (0.4)	4.2 (0.04)	46.6 (4.0)	27.8 (2.6)	26.8 (7.1)	2.2 (0.4)
2010	9233 (551)	11467 (2303)	10633 (1704)	5567 (1724)	5.8 (0.1)	5.9 (0.3)	5.9 (0.3)	4.8 (0.3)	49.8 (6.5)	29.5 (2.9)	28.4 (6.5)	8.0 (1.8)
2011[3]	11533 (833)	n.a	n.a	n.a	5.7 (0.2)	n.a	n.a	n.a	59.2 (7.9)	n.a	n.a	n.a
2012	12167 (850)	n.a	n.a	n.a	5.8 (0.2)	n.a	n.a	n.a	60.4 (6.9)	n.a	n.a	n.a
2013	12700 (500)	n.a	n.a	n.a	5.6 (0.3)	n.a	n.a	n.a	53.2 (3.0)	n.a	n.a	n.a
2014	11533 (1021)	n.a	n.a	n.a	5.5 (0.5)	n.a	n.a	n.a	51.7 (8.4)	n.a	n.a	n.a

[1] The treatments include four distinct management strategies: (1) Abandonment (Ab, selective cutting stoped), indicating areas left unmanaged, (2) Control check (CK, 0%), representing untreated plots; (3) Selective cutting (SC, 15%-30% old), involving the removal of approximately 15%-30% of older bamboos to promote growth of the remaining stand; and (4) Clear-cutting (CC, 100%), where all bamboos are removed, leaving the area completely open. All treatments were implemented in the winter of 2008. Ab was carried out at Sanjiaolun, where the stand was planted 40 years ago. CK, SC, and CC were implemented at Huisun Forest Station, where the stand was planted in an unknown year but is highly likely to have been established in 1985, making it approximately 40 years old. (*Phyllostachys edulis* at both Sanjiaolun and Huisun Forest Station was planted by Dr. Lü Chin-Ming.) Data from Ab were used for Figure 2, while data from CK, SC, and CC were used for Figure 3.

[2] Numbers in parentheses indicate the standard deviation (n = 3).

[3] This *Phyllostachys edulis* plantation had historically suffered from a lack of management and silvicultural care, resulting in a disordered stand structure and a decline in bamboo growth. Broadleaf tree species had also begun to encroach upon the site. In response, Huisun Forest Station undertook a clearance operation in 2011, using excavators to remove the bamboo while retaining larger-diameter broadleaf trees of higher ecological or economic value. Following site preparation, *Liquidambar formosana* seedlings were planted at a planned density of 1,500 stems per hectare, covering a total area of 5 hectares. Early silvicultural treatments, such as weeding, were conducted after planting. Over time, canopy closure has progressively developed within the stand, necessitating thinning interventions to enhance tree growth. Consequently, in 2018, the forest station began planning a thinning operation for the plantation. However, this does not affect our research conclusions. Further details can be found in Chuang et al. (2019).

Table S4. The parameters of allometric models for predicting *P. edulis* biomass with DBH including a, b, and R^2

Culm ages	Sections	$Y = a \times DBH^{b}$		
		a	b	R^2
age 1	Foliage (kg)	0.0046	1.799	0.7908
	Branches (kg)	0.0335	1.6688	0.9429
	Culm (kg)	0.0395	2.4498	0.9946
	Aboveground (kg)	0.0613	2.3308	0.9972
over age 1	Foliage (kg)	0.0049	1.9823	0.2341
	Branches (kg)	0.0426	1.583	0.3379
	Culm (kg)	0.037	2.5909	0.9792
	Aboveground (kg)	0.0652	2.4156	0.9736
General	Foliage (kg)	0.0048	1.9548	0.2854
	Branches (kg)	0.0396	1.612	0.4255
	Culm (kg)	0.0376	2.5632	0.9722
	Aboveground (kg)	0.0637	2.405	0.9672
	Stump (kg)	0.0232	2.3731	0.6833

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